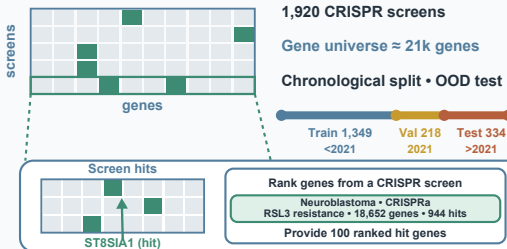
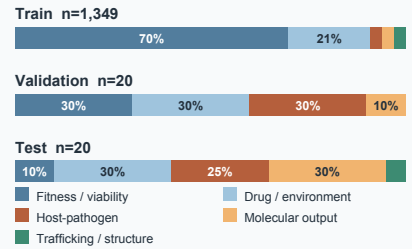


A AssayBench-Loop

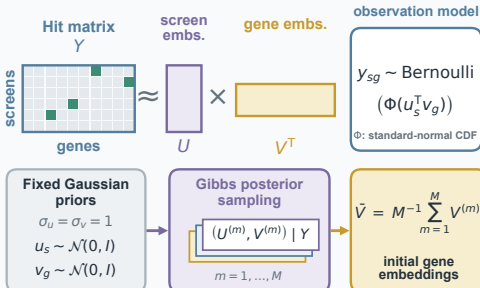
AssayBench data



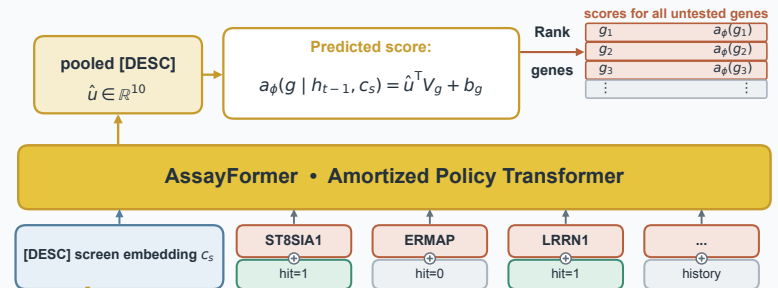
B Phenotype composition



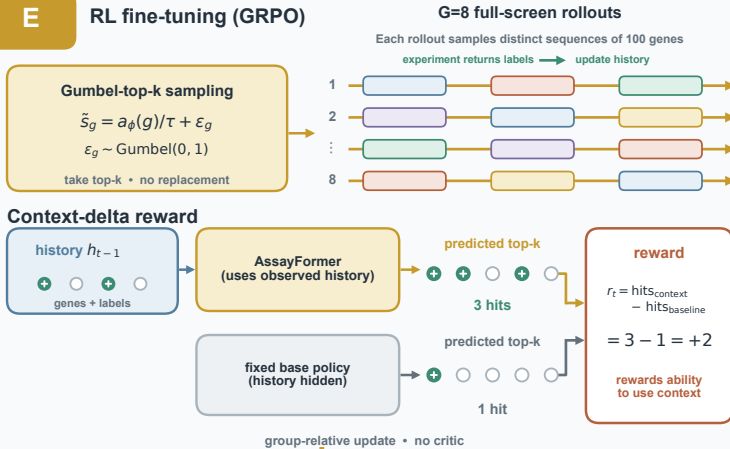
C Embedding Initialization - BPMF



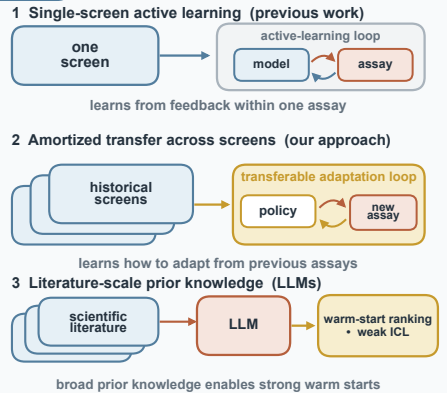
D Amortized policy transformer + supervised objective



E RL fine-tuning (GRPO)

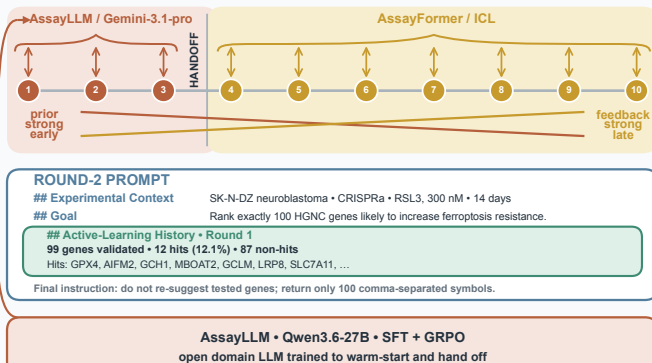


F Three paradigms



G AssayLoop (handoff)

Complementary strengths motivate the handoff
LLM: strong biological prior • AssayFormer: strong in-context learning



H Actual held-out recovery curves

